

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Module 4 | Measures of Spread (4 of 5)

 Reflect and discuss (A)

 1. A survey taken in a large statistics class contained the question: “What’s the fastest you have driven a car (in miles per hour)?” The five-number summary for the 87 males surveyed is:

min = 55, Q1 = 95, Median = 110, Q3 = 120, Max = 155

Should the largest observation in this data set be classified as an _____?

- A. Yes.
- B. No.

 Reflect and discuss (B)

 2. A survey taken of 140 sports fans asked the question: “What is the most you have ever spent for a ticket to a sporting event?”

The five-number summary for the data collected is:

min = 85, Q1 = 130, Median = 145, Q3 = 150, Max = 250

Should the smallest observation be classified as an _____?

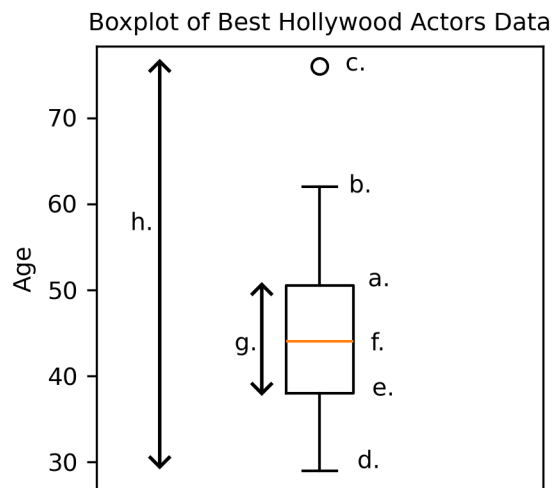
- A. Yes.
- B. No.

Module 4 | Boxplot (2 of 3)

 Reflect and discuss (C)

 3. _____ the points on the boxplot represented by the letters going from a to h to the list below:

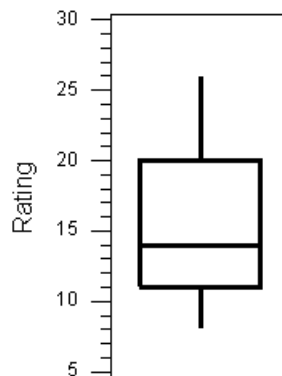
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- A. Q1.
- B. Q3.
- C. M.
- D. Min.
- E. Max.
- F. IQR.
- G. Range.
- H. None of the above.

Reflect and discuss (D)

4. The boxplot below displays ratings for TV shows during sweeps week:



_____ of the shows have a rating greater than which value?

- A. 11.
- B. 15.5.
- C. 14.
- D. 17.

5. What is the percentage of shows that have a rating of less than _____?

- A. 67%.
- B. 75%.
- C. 84%.

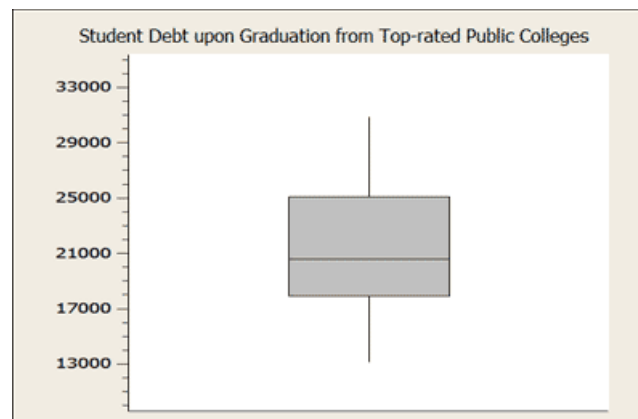
D. 25%.

6. Within which interval of rating would you expect to find the _____ number of shows?

- A. 5-10.
- B. 10-15.
- C. 15-20.
- D. 20-25.
- E. All are equal.

 Reflect and discuss (E)

The Web site www.kiplinger.com published the “Best Values in Public Colleges,” saying they were determined by rankings based on data from more than 500 public four-year colleges and universities. Using out-of-state rankings that considered the academic quality, total out-of-state costs, and average costs after financial aid, the 50 best-rated colleges were selected. The boxplot pictured displays the average debt per graduate for these colleges.



7. What percentage of the graduates will have a debt greater than _____?

- A. 25%.
- B. 50%.
- C. 75%.
- D. 81%.
- E. 19%.

8. _____ of the debts owed are smaller than what amount?

- A. \$18,000.
- B. \$20,600.
- C. \$22,000.
- D. \$25,000.

9. Within which interval of debts would you expect to find the _____ number of graduates?

- A. 13,000–17,000.

- B. 17,000–21,000.
- C. 21,000–25,000.
- D. 25,000–29,000.
- E. All the intervals are equal.

Module 4 | Standard Deviation (1 of 3)

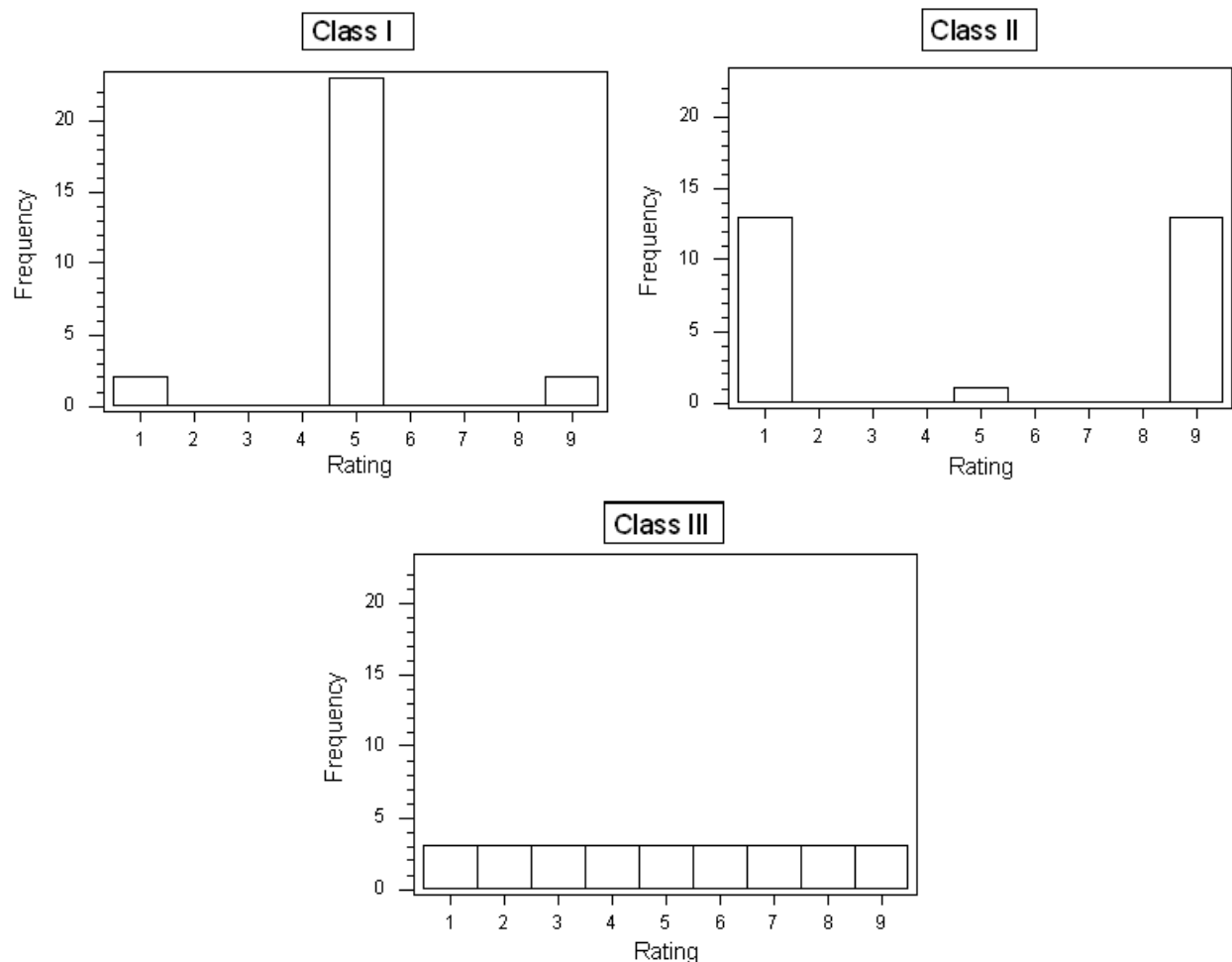
Reflect and discuss (F) Standard Deviation

 10. The concept of standard deviation is less intuitive as a measure of spread than the range or the IQR. The following activity is designed to help you develop a better intuition for the standard deviation.

Background At the end of a statistics course, the 27 students in the class were asked to rate the instructor on a number scale of 1 to 9 (1 being “very poor,” and 9 being “best instructor I’ve ever had”). The following table provides three hypothetical rating data:

Rating	1	2	3	4	5	6	7	8	9
Class I	1	0	0	0	22	0	0	0	1
Class II	12	0	0	0	1	0	0	0	12
Class III	2	2	2	2	2	2	2	2	2

And here are the histograms of the data:



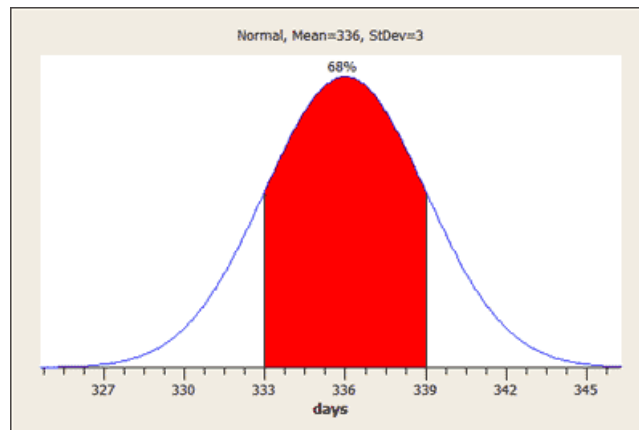
Assume that the average rating in each of the three classes is 5 (which should be visually reasonably clear from the histograms), and recall the interpretation of the SD as a “typical” or “average” distance between the data points

and their mean. Judging from the table and the histograms, which class would have the _____ standard deviation, and which one would have the _____ standard deviation? Explain your reasoning.

Module 4 | Standard Deviation (2 of 3)

Reflect and discuss (G)

Use the Standard Deviation Rule to answer the following questions. This picture of the SD rule applied to this distribution will help:



 11. Almost _____ horse pregnancies fall in what range of lengths?

- A. Above 336 days.
- B. Below 336 days.
- C. Between 333 and 339 days.
- D. Between 330 and 342 days.
- E. Between 327 and 345 days.

 12. What percentage of horse pregnancies last longer than _____ days?

- A. 2.5%.
- B. 16%.
- C. 50%.
- D. 68%.
- E. 100%.

Reflect and discuss (H) Standard Deviation

The body temperatures of dogs are normally distributed with a mean of 101.3°F and a standard deviation of 0.6°F. Use this information to answer the following questions: PracticeProbNum  13. About _____ of the temperatures of dogs fall within which range?

- A. 99.5–103.1.
- B. 100.1–101.9.
- C. 100.1–102.5.
- D. 100.7–101.9.

E. Above 101.3.

 14. What percentage of the temperatures are less than _____?

- A. 0.15%.
- B. 0.3%.
- C. 2.5%.
- D. 5%.
- E. 16%.

 15. What percentage of the temperatures are more than _____?

- A. 0.15%.
- B. 2.5%.
- C. 84%.
- D. 95%.
- E. 97.5%.

Module 5 | The Role-Type Classification (1 of 2)

Reflect and discuss (I)

 16. In example 2 (see corresponding module reading), the number of calories is the { explanatory, response } variable, and the type of hot dog is the { explanatory, response } variable.

 17. In example 4, gender is the { explanatory, response } variable, and smoking habit is the { explanatory, response } variable.

 18. In example 6, age is the { explanatory, response } variable, and legibility distance is the { explanatory, response } variable.

 19. In example 8, favorite type of music is the { explanatory, response } variable, and IQ level is the { explanatory, response } variable.

Module 5 | The Role-Type Classification (2 of 2)

Reflect and discuss (J)

 20. *Example 2:*

_____ is the response variable and it is { categorical, quantitative }.

Type of hot dog is the explanatory variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

 21. *Example 4:*

_____ is the explanatory variable and it is { categorical, quantitative }.

Smoking habits is the response variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

 22. Example 6:

_____ is the explanatory variable and it is { categorical, quantitative }.

Legibility distance is the response variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

 23. Example 8:

_____ is the response variable and it is { categorical, quantitative }.

IQ level is the explanatory variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

 Reflect and discuss (K)

In each of the following three problems, you are presented with a brief description of a study involving two variables. Based on the role-type classification of the two variables, you are asked to determine which of the four cases represents the data structure of the problem.

For your convenience, here again is the role-type classification table:

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

 24. A store asked 250 of its customers whether or not they were satisfied with the service. The purpose of this study was to examine the relationship between the customer's satisfaction and gender.

This study is an example of which _____?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

 25. A study was conducted in order to explore the relationship between the number of beers a person drinks and his/her blood alcohol content (BAC, in %).

This study is an example of which _____?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

 26. A study was conducted in order to determine whether longevity (how long a person lives) is related to a person's handedness (right-handed/left-handed).

This study is an example of which _____?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

Module 5 | Case $C \rightarrow Q$ (2 of 2)

Reflect and discuss (L)

In the following activities, let's explore the Survey of Study Habits and Attitudes (SSHA) to determine if there is there a "gender effect" on SSHA scores.

 27. Do females or males have a higher _____ SSHA score?

- A. Females.
- B. Males.

 28. Do females or males have higher _____ in SSHA scores?

- A. Females.
- B. Males.

Module 5 | Case $C \rightarrow C$ (1 of 2)

Reflect and discuss (M)

 29. The number of _____ in our sample (see corresponding module reading) who feel that they are *underweight* is { 110, 163, 760 }.

 30. The number of _____ in our sample who feel that they are *about right* is { 295, 440, 560, 855 }.

 31. The *total* number of _____ in our sample is { 440, 560, 760, 1200 }.

 32. The *total* number of students who feel they are _____ is { 110, 163, 235, 1200 }.

 Reflect and discuss (N)

 33. Now complete the table by computing the conditional percentages for the _____.

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	$560/760 = 73.7\%$	$163/760 = 21.5\%$	$37/760 = 4.9\%$	$760/760 = 100\%$
	Male	?			100%

 Reflect and discuss (O)

 34. The results (see double bar chart in reading) suggest that the proportion of _____ who are happy with their body weight is { slightly greater, slightly less } than among female students. Females who are not happy with their body weight more often feel they are { about right, overweight, underweight }. Males who are not happy with their body weight feel they are overweight { about as often as, much more often than, much less often than } they feel they are underweight.

 Reflect and discuss (P)

 35. Suppose a study were done to answer the question: “Is the smoking of students related to their parents’ smoking habits?” in which data were collected from 5,375 students and organized in the following two-way table:

		Parents’ Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	1168	3203	4371
	Student Smokes	188	816	1004
	Total	1356	4019	5375

We would like to learn about the relationship between the two categorical variables “parents’ smoking” and “students’ smoking.” More specifically, since we want to explore whether parents’ smoking habits affect their children’s smoking habits, the explanatory variable is { students’ smoking, parents’ smoking } and the response variable is { students’ smoking, parents’ smoking }.

 36. Based on your answer to the question above, which of the following three tables is the appropriate table of conditional percentages with which to supplement the above contingency table “Parent’s Smoking”?

		Table A		
		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	26.7%	73.3%	100%
	Student Smokes	18.7%	81.3%	100%

		Table B	
		Parents' Smoking	
		Parents Do Not Smoke	Parents Smoke
Student Smoking	Student Does Not Smoke	86.1%	79.7%
	Student Smokes	13.9%	20.3%
	Total	100%	100%

		Table C		
		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	21.73%	59.59%	81.32%
	Student Smokes	3.50%	15.18%	18.68%
	Total	25.23%	74.77%	100%

- A. Table A.
- B. Table B.
- C. Table C.

Reflect and discuss (Q)

The United States federal government collects information on Americans who do not have health insurance. Data from 2004 are broken down into 4 regions of the country. These data are summarized in the table provided below. Using this table, answer the following questions.

Region	Uninsured	Insured	Total
Northeast	6,782	47,043	53,825
Midwest	7,757	57,135	64,892
South	19,090	85,800	104,890
West	11,676	55,427	67,103
Total	45,305	245,405	290,710
from 2004 census data			
figures in thousands of people			

 37. We wish to investigate the relationship between which two _____ variables?

- A. Insured and uninsured.
- B. Region and uninsured.
- C. Region and United States.
- D. Health insurance and region.
- E. Health insurance and United States.

 38. We wish to investigate if the region in which one lives affects whether or not one has health insurance.

Which would be the _____ variable?

- A. Health insurance.
- B. Region.